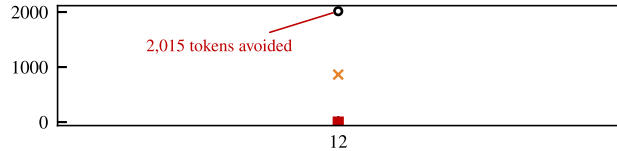


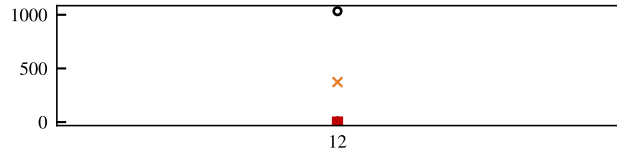
■ AgentTX causal (ours) ···×··· optimistic checkpoint -●- whole-branch abort

Replay tokens



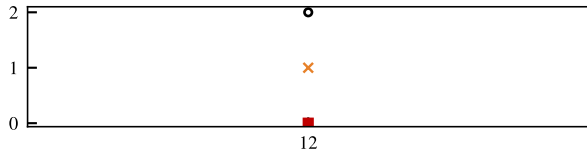
Document size (# entries)
(a) Total LLM work

Completion tokens



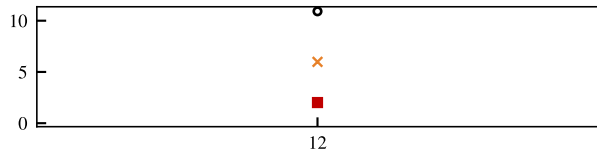
Document size (# entries)
(b) Regenerated content

Documents regenerated



Document size (# entries)
(c) Valid work lost

Recovery p95 (s)



Document size (# entries)
(d) Replay latency